O-S-604D
June 9, 1978
SUPERSEDING
Fed. Spec. O-S-604C
April 9, 1962

FEDERAL SPECIFICATION

SODIUM METASILICATE, TECHNICAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of sodium metasilicate (see 6.1).

1.2 Classification. Sodium metasilicate shall be of the following types, as specified (see 6.2):

Type I - Pentahydrate ($Na_2SiO_3 5H_2O$).
Type II - Anhydrous (Na_2SiO_3).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-D-705 - Drum, Shipping and Storage: Steel, 16 and 30 Gallon Capacity.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6850

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3. REQUIREMENTS

3.1 Appearance. Sodium metasilicate shall be white and granular.

3.2 Physical and chemical characteristics. Sodium metasilicate shall conform to the applicable physical and chemical characteristics of table I when tested as specified in 4.2.2.

TABLE I. Physical and chemical characteristics

Characteristics	Type I (percent by wt.)	Type II (percent by wt.)
Total alkalinity (as Na_2O)	28.5 to 30.0	49.5 to 53.5
Total silica (as SiO_2)	27.5 to 29.0	44.0 to 49.2
Insoluble matter, maximum	0.2	0.25
Material passing a 149-micron (No. 100) sieve, maximum	10.0	3.0
Material retained on a 2.00 mm (No. 10) sieve, maximum	10.0	10 0

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Quality conformance inspection.

4.2.1 Examination of preparation for delivery. Sampling for inspection shall be conducted in accordance with MIL-STD-105, inspection level S-2, AQL 4.0 percent defective. An examination shall be made to determine compliance with the requirements of section 5.

4.2.2 Testing. Sample size shall be one pound, drawn at random from the lot. Testing shall be conducted as specified in 4.3. The lot shall be rejected if the sample fails one test.

4.3 Test methods. Tests shall be conducted as follows:

4.3.1 Total alkalinity. Total alkalinity as sodium oxide (Na_2O) shall be determined as follows:

Weigh 20 g of the sample to the nearest mg, transfer to a 500-ml volumetric flask and dilute to the mark. Transfer a 50-ml aliquot to a 250-ml beaker and titrate with 0.5N-hydrochloric acid (HCl) using methyl orange as the indicator. Reserve the titrated solution for the determination of total silica. Calculated the total alkalinity by:

$$\text{Total alkalinity as } Na_2O, \text{ percent} = \frac{N \times V \times 3.1}{W}$$

where: N = normality of the HCl
V = milliliters of HCl required for titration, and
W = grams of sample in the aliquot.

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4.3.2 Total silica. Total silica as silicon dioxide (SiO_2) shall be determined as follows. transfer the titrated solution as obtained in 4.3.1 to a porcelain evaporating dish, add 25 ml of HCl (sp gr 1.19), and evaporate to apparent dryness on a steam bath. Triturate the dehydrated residue with the smooth end of a stirring rod, moisten the residue with 10 ml of HCl (1 + 1), and again evaporate to apparent dryness on the steam bath. Dehydrate at 110C for 1 h, take up the residue with 10 ml of HCl (1 + 1) and 20 ml of water, and digest a short time on the steam bath to effect solution of the soluble salts. Filter the silica on a fine-texture paper by washing the dish with hot water. Scrub the dish with a rubber policeman and again wash thoroughly with hot water. Wash the residue and paper free of acid with hot water and reserve.

Evaporate the filtrate and washings on the steam bath in the porcelain dish used before, moisten the residue with 10 ml of HCl (1 + 1), and again evaporate to dryness. Dehydrate at 110 C for 1 h, take up the residue with 10 ml of HCl (1 + 1) and 20 ml of water, digest as before to dissolve soluble salts, and filter off any additional silica on a separate filter paper. Scrub the dish and wash the residue and filter paper free from acid as before.

Transfer both papers and residues to a platinum crucible previously ignited and weighed without cover, and ignite in a muffle furnace until free from carbon, heating slowly at first. Cover the crucible with a platinum cover, heat to the highest temperature of a blast lamp for 15 mm, cool in a desiccator, and weigh without the crucible cover.

Add 5 ml of water to the contents of the crucible and 2 or 3 drops of H_2SO_4 (1 + 1), then slowly introduce approximately 10 ml of HF (sp gr 1.15). Evaporate to a small volume on the steam bath, add another portion of about 10 ml of HF, and evaporate to fumes of H_2SO_4. Heat the crucible, gently at first, over an open flame to drive off H_2SO_4, and finally at a bright red heat. Cool in a desiccator, and weigh. The loss in weight represents SiO_2.

Calculate the percentage of silica (SiO_2) as follows:

$$SiO_2, \text{ percent} = [(A - B)/W] \times 100$$

where A = grams of ignited residue before treatment with HF
B = grams of ignited residue after treatment with HF, and
W = grams of sample in aliquot.

4.3.3 Insoluble matter. Matter insoluble in water shall be determined as follows:

Weigh 100 g of the sample to the nearest 0.5 g and transfer to a 1-liter beaker. Dissolve by stirring with water at room temperature and dilute to approximately 900 ml. Filter by suction through a prepared, dried, and weighed Gooch crucible, using on the crucible a pad made of asbestos fiber only. Wash the beaker and residue free from alkali with water, and dry the crucible to constant weight in an oven at 110 C. Cool in a desiccator, and weigh.

Calculate the percentage of matter insoluble in water from the average gain in weight of two checking duplicate determinations as follows

Matter insoluble in water, percent = grams of residue

4.3.4 Particle size. Particle size shall be determined as follows:

Use a nest of clean, dry sieves consisting of a 2-mm (top) and a 149-micron sieve. Transfer 100 ± 0.1 g sample to the top sieve Simultaneously shake the sieves with occasional tapping of the edge of the bottom sieve on a large, thick piece of rubber. When the portion of the sample passing through the bottom sieve is less than 0.1 g/min of shaking. (This can be well judged by sifting into a large, dark colored pan.) Stop shaking and weigh the material retained on the top sieve and the material in the catch pan.

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5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. One-hundred ± 1/2 pound of sodium metasilicate shall be packed in a type III or IV steel drum of PPP-D-705.

5.1.2 Commercial. The sodium metasilicate shall be packed in containers to insure delivery at destination, and to provide for redistribution by the initial receiving activity.

5.2 Palletization. When shipments to Government depots are full car or truckload, the shipping containers shall be palletized on wooden pallets, 2 or 4 way design, to facilitate handling, in accordance with normal commercial practice. The palletized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.3 Marking. Marking shall be as specified in the contract or order (see 6.2).

6. NOTES

6.1 Intended use. Sodium metasilicate is intended for use in various washing, cleaning, and scouring processes, with or without soap or detergents as conditions demand.

6.1.1 Use concentration. In normal usage 1 lb of type II sodium metasilicate is equivalent in washing efficiency to 1.6 lb of type I.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of material required (see 1.2).
(c) Packing required (see 5.1).
(d) Marking required (see 5.3).

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E 29).

6.5 Military use. Items procured under this specification for military use are to be limited to the variety shown in the applicable military standard or military supply standard. Personnel of the military departments are requested to refer to these documents for guidance.

MILITARY INTEREST:

Custodians:

Army - EA
Air Force - 68

Review activities:

Army-EA, GL, MD
Air Force - 68
DSA-GS

User activities:

Army-MI, SM
Navy - OS

Civil Agency Coordinating Activities:

DOT-ACO
GSA-FSS, PBO
VA-DMS

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 50 cents each.

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